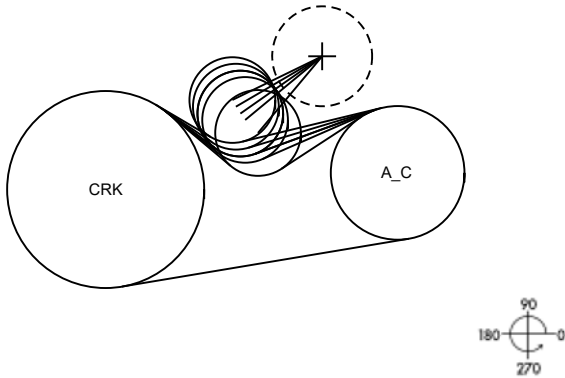




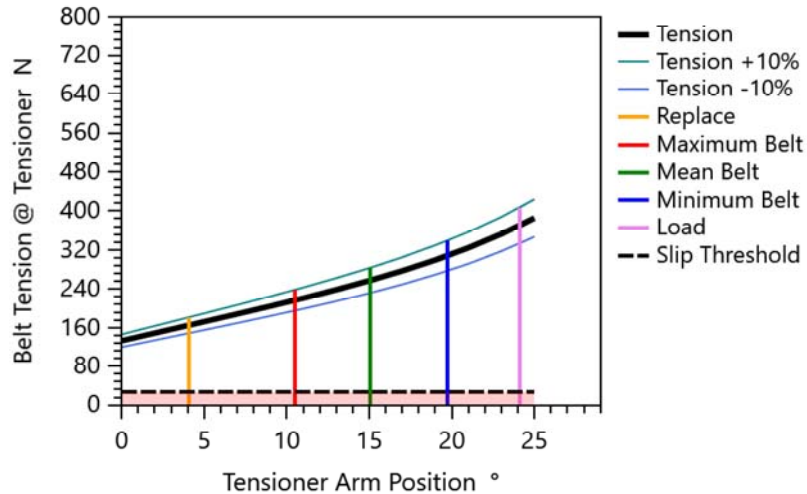
**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

**Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016**

|                                     |               |
|-------------------------------------|---------------|
| Belt Code                           | MT610         |
| # of Ribs / Cord Material           | 4 / polyester |
| Effective Belt Length mm (ISO 9981) | 1013          |
| <b>Tensioner Data</b>               |               |
| Type                                | Automatic     |
| Design Tension N                    | 258 (58 lbf)  |



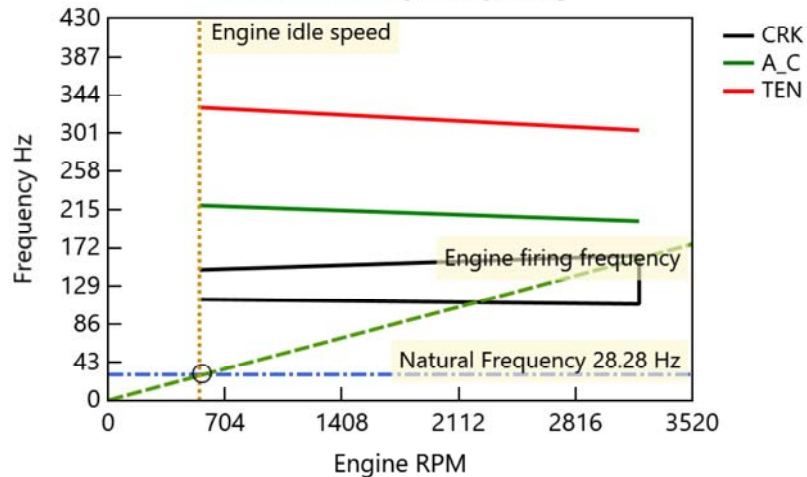
## Belt Tension Control



## Belt Life B10

| Duty Cycle<br>DCn-th % | Av. Vehicle<br>Speed km/hr | PK-2_2p-MT3<br>hours |
|------------------------|----------------------------|----------------------|
| 95                     | 1                          | 15894                |

## Natural Frequency Map



## Peak Tension & Hubload

|     | Engine<br>RPM | Accel.<br>RPM/s | Tension<br>N | Hubload<br>N | Direction<br>/Wrap ° |
|-----|---------------|-----------------|--------------|--------------|----------------------|
| CRK | 1400          | 1000            | 626          | 826.0        | 356 / 226            |
| A_C | 1400          | 1000            | 264          | 886.9        | 193 / 191            |
| TEN | 550           | 1000            | 258          | 248.4        | 83 / 57              |

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

## Permissible Grooved Pulley

### Axial Offset Allowance

For 0.90° Belt Entry Angle

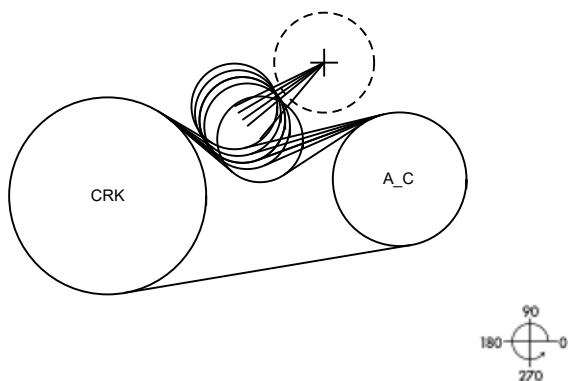
| Adjacent<br>Grooved<br>Pulleys | Flat<br>Pulley<br>Angular<br>Misalignment<br>° | Axial<br>Offset<br>Allowance<br>mm | Grooved<br>Pulley<br>Axial<br>Offset<br>% | Attributed to<br>Flat Pulley<br>Angular<br>Misalignment<br>% |
|--------------------------------|------------------------------------------------|------------------------------------|-------------------------------------------|--------------------------------------------------------------|
| A_C->CRK                       |                                                | 4.11                               | 100                                       |                                                              |
| CRK->A_C                       | TEN (1.20)                                     | 0.94                               | 11                                        | 89                                                           |



**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

## Layout Data mm

|     | X      | Y     | Flat  | Pitch  | Effective | DOB    |
|-----|--------|-------|-------|--------|-----------|--------|
| CRK | 0.00   | 0.00  |       | 177.41 | 175.01    | 176.00 |
| A_C | 263.00 | 15.00 |       | 120.41 | 118.01    | 119.00 |
| TEN | 121.28 | 68.38 | 75.00 | 77.20  | 79.60     |        |



## Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

|                                        |               |
|----------------------------------------|---------------|
| Belt Code                              | MT610         |
| # of Ribs / Cord Material              | 4 / polyester |
| Stretch and Wear Allowance % of Length | 0.70          |
| Belt Length Tolerance (+/-) mm         | 5.00          |

## Tensioner Data

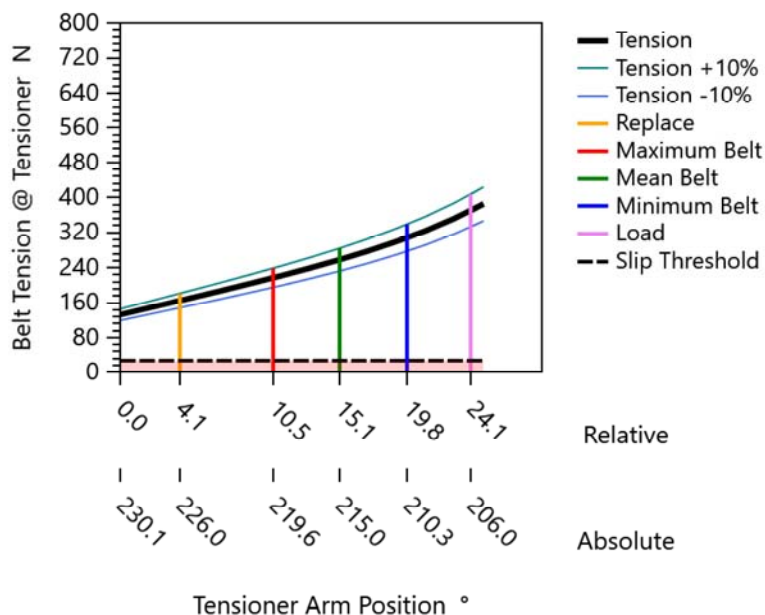
|                                   |                 |
|-----------------------------------|-----------------|
| Type                              | Automatic       |
| Design Tension N                  | 258 (58 lbf)    |
| Pivot Point {X, Y Coordinates} mm | 195.00 , 120.00 |
| Arm Length mm                     | 90.0            |
| Spring Mean Load Nm               | 16.07           |
| Spring Pre-Load Nm                | 8.46            |
| Spring Rate Nm/deg                | 0.505           |

## RESULTS

### Belt Drive System Geometry

|     | Span Length<br>mm | Wrap Angle<br>° | Speed Ratio<br>(Ref. Engine) |
|-----|-------------------|-----------------|------------------------------|
| CRK | 261.9             | 226.2           | 1.000                        |
| A_C | 114.8             | 190.6           | 1.473                        |
| TEN | 56.4              | 56.8            | 2.298                        |

### Belt Tension Control





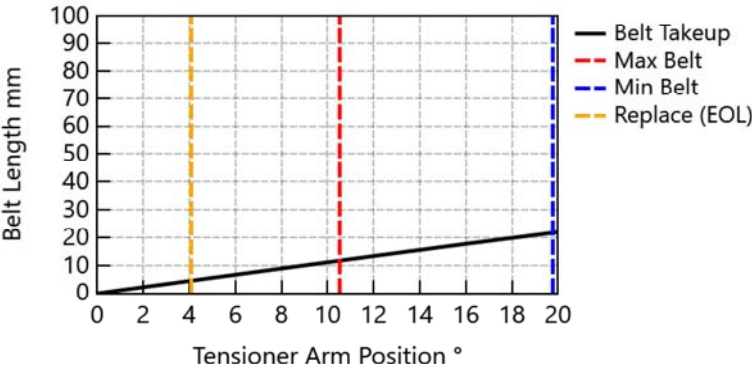
AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022

Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/° 1.087

| Position Description         | Belt-Drive Length |         |        |        |        |        |
|------------------------------|-------------------|---------|--------|--------|--------|--------|
|                              | Free Arm          | Replace | Max    | Mean   | Min    | Load   |
| Arm Position °               | 230.1             | 226.0   | 219.6  | 215.0  | 210.3  | 206.0  |
| Idler X Coordinate mm        | 137.2             | 132.5   | 125.6  | 121.3  | 117.3  | 114.1  |
| Idler Y Coordinate mm        | 51.0              | 55.3    | 62.7   | 68.4   | 74.6   | 80.6   |
| Hubload to Arm Angle °       | 36.5              | 38.7    | 43.0   | 46.7   | 51.1   | 55.6   |
| Hubload Direction °          | 86.5              | 84.7    | 82.6   | 81.7   | 81.4   | 81.6   |
| Hubload N                    | 158.2             | 187.2   | 224.4  | 245.2  | 263.4  | 277.9  |
| Tension N                    | 132.8             | 165.2   | 217.2  | 257.8  | 308.6  | 370.4  |
| Idler Wrap Angle °           | 73.1              | 69.0    | 62.2   | 56.8   | 50.5   | 44.1   |
| Effective Drive Length mm    | 1030.9            | 1026.3  | 1019.2 | 1014.2 | 1009.2 | 1004.8 |
| Required Eff. Belt Length mm | 1030.1            | 1025.6  | 1018.4 | 1013.4 | 1008.4 | 1004.0 |

Belt Take-up

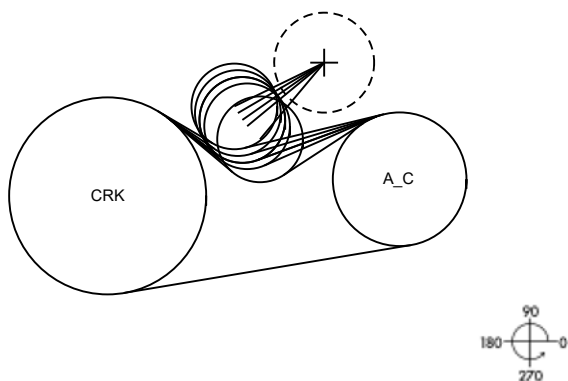




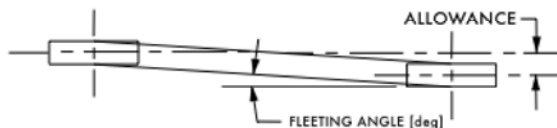
**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

## Alignment Data °

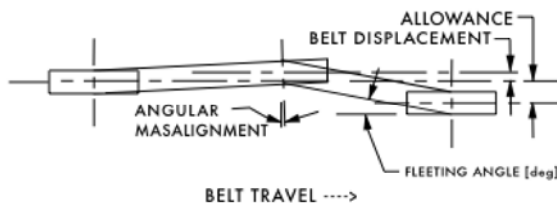
|                          |      |
|--------------------------|------|
| Fleeting Angle           | 0.90 |
| TEN Angular Misalignment | 1.20 |



## (GROOVED - GROOVED) PULLEYS



## (GROOVED - FLAT - GROOVED) PULLEYS



## RESULTS

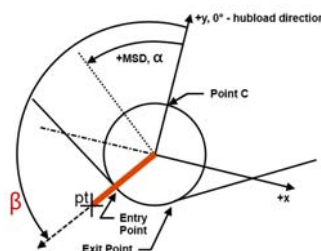
### Permissible Grooved Pulley

#### Axial Offset Allowance

For 0.90° Belt Entry Angle

| Adjacent Grooved Pulleys | Flat Pulley Angular Misalignment ° | Axial Offset Allowance mm | Grooved Pulley Axial Offset % | Attributed to Flat Pulley Angular Misalignment % |
|--------------------------|------------------------------------|---------------------------|-------------------------------|--------------------------------------------------|
| A_C->CRK                 |                                    | 4.11                      | 100                           |                                                  |
| CRK->A_C                 | TEN (1.20)                         | 0.94                      | 11                            | 89                                               |

### Hubload to Arm Angle & MSD [°]



Hubload-to-arm angle is the angle between hubload and arm where arm direction is from pulley center to pivot.

$\beta$  = Hubload-to-arm angle  
pt = Tensioner pivot point

### Backside Pulley Alignment Sensitivity

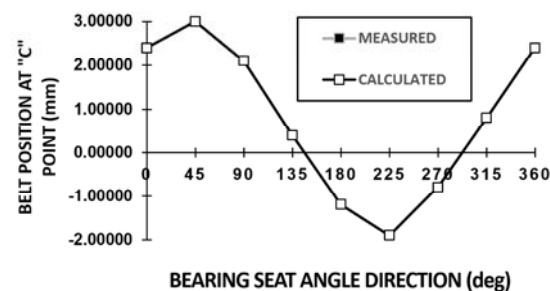
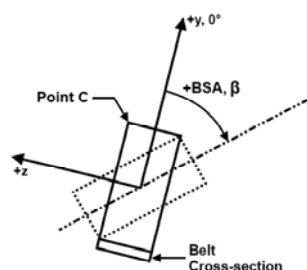
| Backside Pulley | Angular Misalignment | Most Sensitive Direction | Resulting Fleeting Angle | Suggestions |
|-----------------|----------------------|--------------------------|--------------------------|-------------|
| 3: TEN          | 1.200                | 75.95                    | 0.80                     | AB C E      |

Suggestion List for Backside Pulley Alignment Sensitivity

- A] Increase EXIT span length
- B] Decrease ENTRANCE span length
- C] Increase both ENTRANCE / EXIT span lengths
- D] Increase wrap angle
- E] Decrease wrap angle

Note: Most Sensitive Direction for Grooved pulleys is parallel to the belt entry span.

### Direction of Bearing Seat Angle [°]

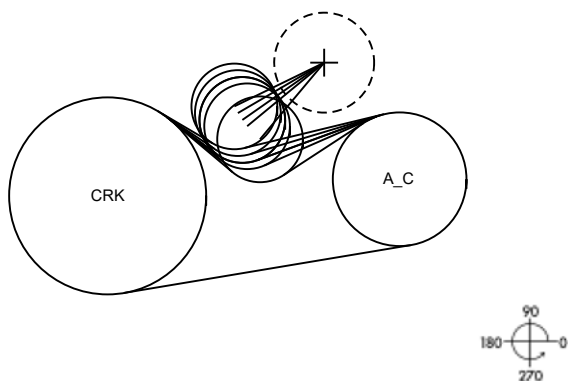
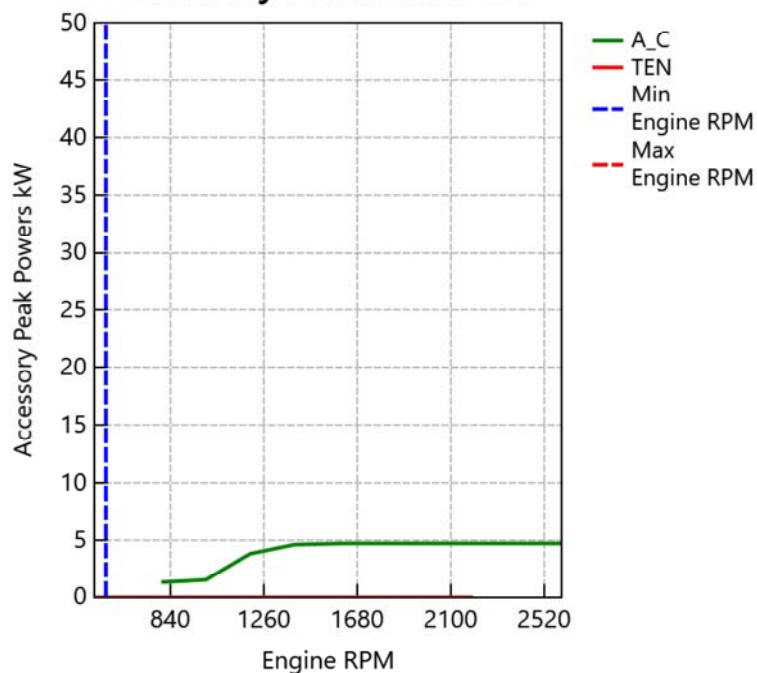




**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

|                           |                |
|---------------------------|----------------|
| Design Tension N          | 258 (58 [lbf]) |
| Engine Acceleration RPM/s | 1000           |
| Engine Deceleration RPM/s | 1000           |

## Accessory Peak Powers kW



## Accessory Data

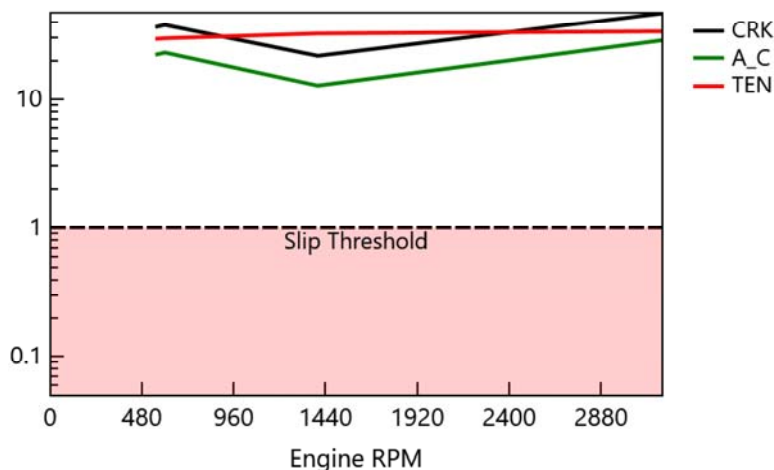
|     | File Name - if used               | Moment of Inertia kg m^2 |
|-----|-----------------------------------|--------------------------|
| A_C | SD7H15-AC.cmp                     | 0.0034                   |
| TEN | MX14D_MY27_ALT-24V-150Amp-EU-F... | 0.0009                   |

## RESULTS

### Belt Slip Sensitivity

| Pulley | Most Critical Load Condition |                    |     |  |  | Required Increase<br>Tension or Wrap Angle |   |
|--------|------------------------------|--------------------|-----|--|--|--------------------------------------------|---|
|        | Acceler.<br>RPM/s            | Load Combination % | A_C |  |  | N                                          | ° |
| CRK    | 1000                         | 100                |     |  |  |                                            |   |
| A_C    | 0                            | 100                |     |  |  |                                            |   |
| TEN    | 1000                         | 100                |     |  |  |                                            |   |

### Belt Slip Safety Factor





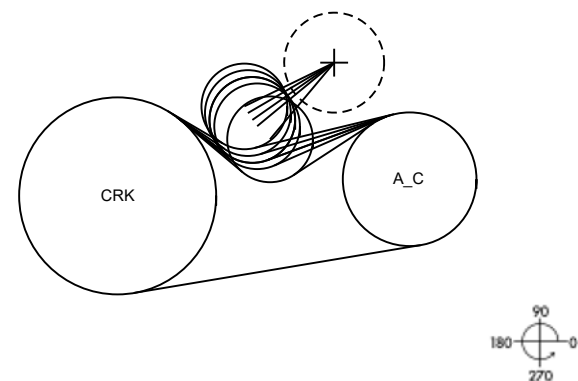
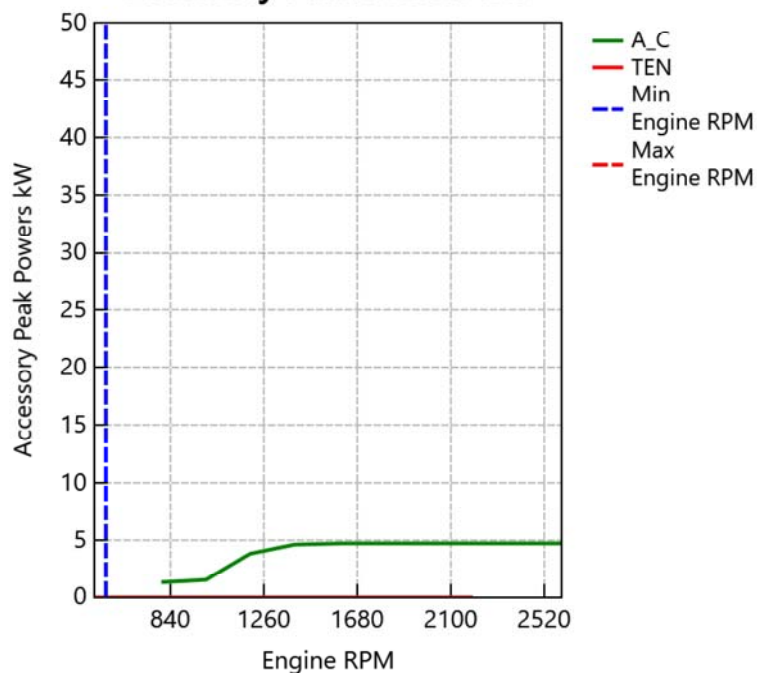
User: Abdurrahman Kol

Design: AG0868\_BMC-Power AC-SD7H15\_Drive 4PK1013HD\_E9843-16Nm\_27Dec2022

**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

|                           |                |
|---------------------------|----------------|
| Design Tension N          | 258 (58 [lbf]) |
| Engine Acceleration RPM/s | 1000           |
| Engine Deceleration RPM/s | 1000           |

## Accessory Peak Powers kW



## RESULTS

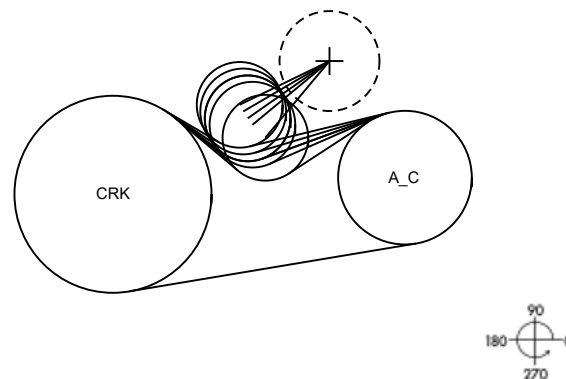
### Peak Tension & Hubload

|     | Engine RPM | Accel. RPM/s | Tension N | Hubload N | Direction /Wrap ° | Shear Stress N/mm-rib @ RPM |
|-----|------------|--------------|-----------|-----------|-------------------|-----------------------------|
| CRK | 1400       | 1000         | 626       | 826.0     | 356 / 226         | 0.270 @ 1400                |
| A_C | 1400       | 1000         | 264       | 886.9     | 193 / 191         | 0.471 @ 1400                |
| TEN | 550        | 1000         | 258       | 248.4     | 83 / 57           | N/A                         |

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)



**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**



**Load Conditions kW for DC 95%**

| %     | Eng.<br>RPM | V,<br>km/hr | T,<br>°C | CRK  | A_C  | TEN  |
|-------|-------------|-------------|----------|------|------|------|
| 27.00 | 800         | 1           | 90       | 1.31 | 1.30 | 0.01 |
| 10.00 | 1000        | 1           | 90       | 1.51 | 1.50 | 0.01 |
| 13.00 | 1250        | 1           | 90       | 4.01 | 4.00 | 0.01 |
| 18.00 | 1500        | 1           | 90       | 4.61 | 4.60 | 0.01 |
| 19.00 | 1750        | 1           | 90       | 4.71 | 4.70 | 0.01 |
| 13.00 | 2000        | 1           | 90       | 4.71 | 4.70 | 0.01 |

**Mean Hubloads N for DC 95%**

| %     | Eng<br>RPM | CRK | A_C | TEN |
|-------|------------|-----|-----|-----|
| 27.00 | 800        | 640 | 690 | 246 |
| 10.00 | 1000       | 627 | 676 | 246 |
| 13.00 | 1250       | 803 | 859 | 246 |
| 18.00 | 1500       | 789 | 844 | 245 |
| 19.00 | 1750       | 749 | 803 | 245 |
| 13.00 | 2000       | 714 | 767 | 245 |

**Mean Tensions N for DC 95%**

| %     | Eng<br>RPM | CRK | A_C | TEN |
|-------|------------|-----|-----|-----|
| 27.00 | 800        | 434 | 259 | 258 |
| 10.00 | 1000       | 420 | 259 | 258 |
| 13.00 | 1250       | 603 | 259 | 258 |
| 18.00 | 1500       | 589 | 258 | 258 |
| 19.00 | 1750       | 547 | 258 | 258 |
| 13.00 | 2000       | 511 | 258 | 258 |

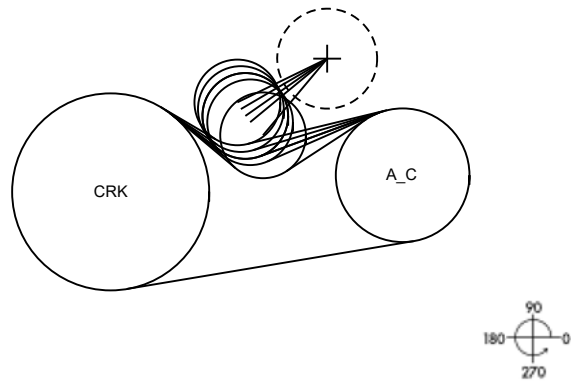




AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022

|                                                  |               |
|--------------------------------------------------|---------------|
| Load Conditions [kW]                             |               |
| Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016 |               |
| Belt Code                                        | MT610         |
| # of Ribs / Cord Material                        | 4 / polyester |

|                  |              |
|------------------|--------------|
| Tensioner Data   |              |
| Type             | Automatic    |
| Design Tension N | 258 (58 lbf) |







**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

## RESULTS

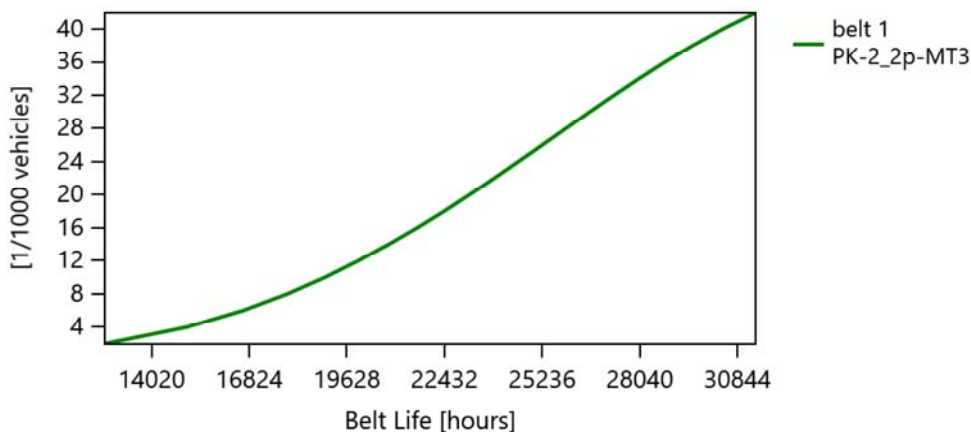
### Pulley Contributions to Belt Rib Fatigue %

|             | CRK | A_C  | TEN  |
|-------------|-----|------|------|
| PK-2_2p-MT3 | 2.7 | 25.0 | 72.4 |

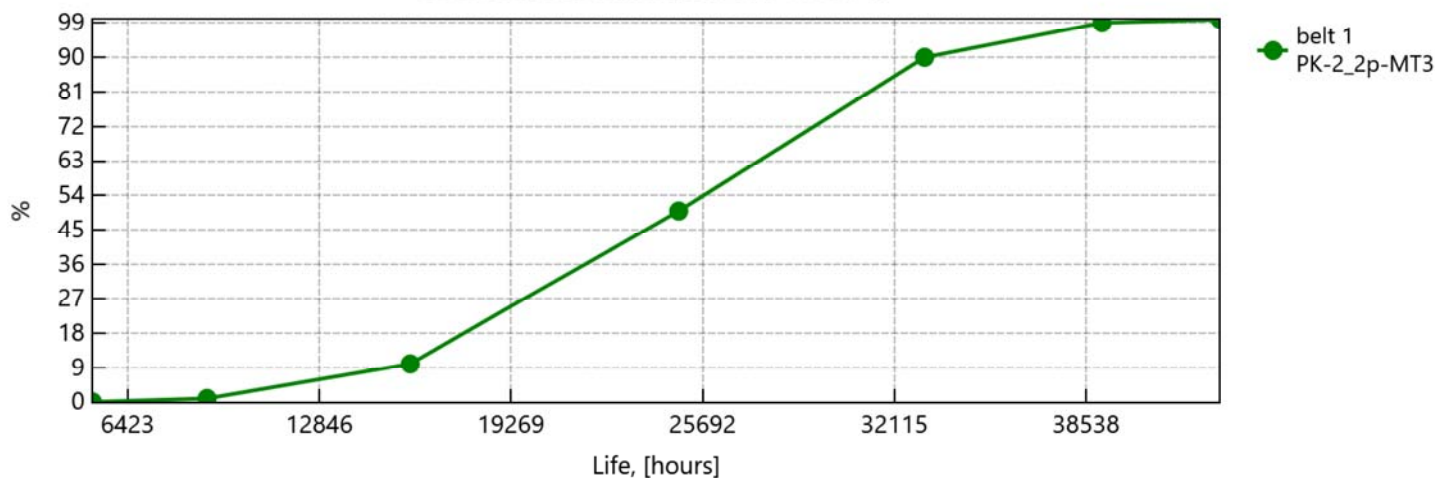
### Belt Life B10

| Duty Cycle<br>DCn-th % | Av. Vehicle<br>Speed km/hr | PK-2_2p-MT3<br>hours |
|------------------------|----------------------------|----------------------|
| 95                     | 1                          | 15894                |

### Belt Incidents per Thousand Vehicles



### Weibull Distribution for DC95%



### Load Conditions kW for DC 95%

| %     | Eng.<br>RPM | V,<br>km/hr | T,<br>°C | CRK  | A_C  | TEN  |
|-------|-------------|-------------|----------|------|------|------|
| 27.00 | 800         | 1           | 90       | 1.31 | 1.30 | 0.01 |
| 10.00 | 1000        | 1           | 90       | 1.51 | 1.50 | 0.01 |
| 13.00 | 1250        | 1           | 90       | 4.01 | 4.00 | 0.01 |
| 18.00 | 1500        | 1           | 90       | 4.61 | 4.60 | 0.01 |
| 19.00 | 1750        | 1           | 90       | 4.71 | 4.70 | 0.01 |
| 13.00 | 2000        | 1           | 90       | 4.71 | 4.70 | 0.01 |

### Load Contribution to Belt Rib Fatigue for DC 95%

| %     | Vblt<br>m/s | CRK<br>Tt N | CRK<br>Ts N | %<br>PK-2_2p-MT3 |
|-------|-------------|-------------|-------------|------------------|
| 27.00 | 7.4         | 434         | 258         | 14               |
| 10.00 | 9.3         | 420         | 258         | 6                |
| 13.00 | 11.6        | 603         | 258         | 14               |
| 18.00 | 13.9        | 589         | 258         | 23               |
| 19.00 | 16.3        | 547         | 258         | 25               |
| 13.00 | 18.6        | 511         | 258         | 18               |



User: Abdurrahman Kol

Design: AG0868\_BMC-Power AC-SD7H15\_Drive 4PK1013HD E9843-16Nm 27Dec2022

**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

## Belt Data, DB Ver. = 2.13, Belt Rev. = 7/14/2016

|                           |               |
|---------------------------|---------------|
| Belt Code                 | MT610         |
| # of Ribs / Cord Material | 4 / polyester |

## Tensioner Data

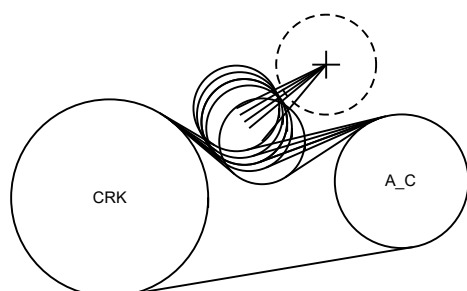
|                                                |              |
|------------------------------------------------|--------------|
| Type                                           | Automatic    |
| Design Tension N                               | 258 (58 lbf) |
| Inertia of the Tensioner Arm Kg-m <sup>2</sup> | 0.0009       |
| Tensioner Pulley Mass kg                       | 0.80         |

## Engine Data

|                                                |        |
|------------------------------------------------|--------|
| # of Cylinders                                 | 6      |
| Crankshaft Moment of Inertia kg m <sup>2</sup> | 0.5000 |

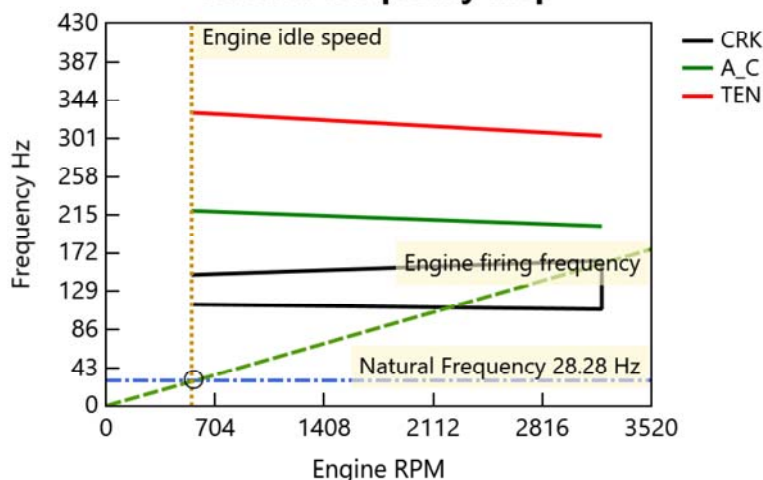
## Accessory Data

|     | File Name - if used              | Moment of Inertia kg m <sup>2</sup> |
|-----|----------------------------------|-------------------------------------|
| A_C | SD7H15-AC.cmp                    | 0.0034                              |
| TEN | MX14D_MY27_ALT-24V-150Amp-EU-... | 0.0009                              |



## RESULTS

### Natural Frequency Map



Note: Only low frequency spans are shown.  
System Resonance (Mode 1: 28.28)

Note: If Engine Idle is near system resonant RPM,  
then accessory drive system may have  
excessive vibration.  
If system resonance is within a span  
frequency window, then span vibration  
may have excessive amplitude.



**AG0868 BMC-Otomotif- 6Cyln.  
Secondary AC (SD7H15) Drive  
Tensioner; E9843/16Nm@15°  
Tensioner Position 195/120  
4PK1013HD Polyester; 27Dec2022**

**Drive Notes**

| Date | Note |
|------|------|
|------|------|